KEDGE dart v32: build day run sheet

Dart: Ø42 x 152 mm, 1.5 mm walls, PLA on the X2D. Kick leg 0 to 200.75°, drill pair 0 to 45.0°. Collision sweep clean (kick in 202 steps to 200.75°, drills in 1° steps to 45.0°, every sub-part, exact meshes). Printed volume 112.07 cm³ (about 139.0 g if solid; less at 40% infill). Hard stops checked: pods stop on the hull horns about 1° past 45.0° and on the fold lugs, the kick on its rest 1.5° past the plant. Launcher fit checked against the real launcher CAD. About 12 hours wall clock with the two plates printing in the background.
Rules for the whole day: the power loop plug stays OUT while you solder, move a wire or plug in USB. Never let the two cells touch metal together. One step at a time; each step says what you should see before you move on.

1. What is in the pack

File	What it is
print/plate_A_hull.3mf	Print plate A: hull top, hull bottom, nose, tail cap, both pod caps (PLA). Plate C in PPA-CF: both couplers, the tail boom and its anchor disc. Already in print orientation.
print/plate_B_legs_tray.3mf	Print plate B: tray, kick leg, both drill pods, three augers. Already in print orientation.
print/fit_coupon.stl	20 minute test print: checks every fit before the long plates (section 4).
print/*.stl, step/*.step	Every printed part on its own, oriented for printing. STEP is the editable CAD (augers are faceted in STEP).
docs/wire_cut_list.csv	All 50 wires (firmware v4 harness, DRV8833 #3 off the dart), pad to pad, colour, gauge, cut length on the v31 routes (also section 9).
firmware/dart/ (repository)	Firmware v4 and its WIRING.md (the soldering order) and BENCH_TEST.md. Follow WIRING.md Path B for the harness; this sheet covers the printing and the mechanical build.
firmware/test/	PC simulation of the control logic (all tests pass).
docs/KEDGE_*_engineering_report.pdf	Flight, launch and landing loads, holding force by soil, reel limit, launcher silhouette, design versions.
launcher/barrel_profile.dxf / barrel_keepout.stl	What the launcher barrel has to clear, with 1 mm clearance.
source/	The parametric generator and the whole check pipeline (run_all.sh). Every number lives in P at the top; KOV overrides make variants.

2. Parts and tools to have on the bench

Item	Qty	Note
XIAO ESP32S3 (+ its FPC antenna), DRV8833 x2, INA219, BNO085, MP1584 (the 5.04 V one)	1 each	Headers clipped off (flush cutters), INA219 screw terminal cut off. DRV8833 #3 goes to the base station. MP1584 #2 stays spare.
OTOTEC 2S BMS stick	1	Caliper it when it arrives: the pocket under the cells is 40.8 x 8 x 2.1 plus 0.5 each way.
18350 cells, charged	2	Charge both to 4.2 V in the XTAR first.
Worm N20 (kick, deploy) + third worm N20 (foot, buy)	3	Buy the foot one single-shaft if listed; otherwise cut its lower stub to 2 mm (it would hit the hull top when folded). Cut the kick one's RIGHT stub (the one away from the leg) to 1 mm past the worm box: it stays inside the saddle pocket.
Drill motors (12 x 10 x 24.27, M3 x 56 shaft)	2	Each sits INSIDE its auger sleeve, held by its can in the arm's bushing. Bend both solder tabs back along the can before soldering: 1.2 mm behind the end cap.
F693ZZ flanged bearing	1	In the kick ear (the drills need none now).
M3 nuts / M3 nylocs	2 / 2	One jammed pair per drill sleeve: nut in the hub's side slot, nyloc on it through the tip.
M3 x 8 countersunk screw (one more of the radial ones)	1	The tether pin in the tail boom eye (head flush, threads into the lower tine; threadlocker).
M3 x 8 screws	8 + 2	8 countersunk (flat head, DIN 7991) radial screws flush with the skin: 4 into tray disc 1, 4 through the tail spigot into the boom anchor; 2 pan head through the printed pod caps into the couplers.
M3 x 8 screw (kick hub)	1	Through the ear bearing into the hub (its head clamps the bearing's inner race onto the hub boss: no washer).
M3 grub screw	1	Foot auger on the foot N20 shaft flat.
2xAA battery holders (for their contacts)	2	You need 2 springs and 2 flat plates. Launch throws the cells backwards, so springs go at the FRONT of the cell bay and flat plates at the BACK.
470 to 1000 µF capacitor, 10 V or more	1	Across the MP1584 5 V output: rides through contact bounce on landing.
Schottky diode 1N5817 or SS14	1	In wire 18 (5 V to the XIAO): stops back-feeding USB.

Item	Qty	Note
JST-XH 3-pin pair, JST-XH 2-pin pair + a 2-pin shorting plug	1 each	Charge socket in the bottom seat of the tail cap, power loop socket in the TOP seat (the shorting plug sits recessed under the skin, a short wire loop on it is the pull tab).
Silicone wire 22, 24, 26, 28 AWG		Totals per colour in section 9.
Zip ties 2.5 mm (2: deploy N20 cradle, foot N20 strap), medium threadlocker (or CA), double-sided foam tape, hot glue		No nose ballast in v31: at the real 7 m/s launch it adds nothing (dart-physics).
Already on your bench		Caddx Ant camera, Zeus VTX + u.FL antenna, 3S LiPo + XT60 lead #1, XTAR MC2 (charges the 18350s out of the dart), breadboard and jumpers.
Tools		Soldering iron, solder, flux, multimeter, wire strippers, flush cutters, small flat screwdriver (MP1584 trimmer), heat shrink + lighter or heat gun, JST-XH crimper or pre-crimped pigtails, 5-minute epoxy, hex keys, tweezers, luggage scale (pull test).

3. Timeline

Clock	Step	Printer
0:00	4. Fit coupon	coupon (20 min)
0:30	5. Start plate A, flash the firmware	plate A (about 3 h)
1:00	6. Prepare boards, set both MP1584s, cut and label all wires	plate A
2:00	7. Solder the bench harness, power-up test, motor directions	plate A
3:30	Start plate B (and the two couplers in PPA-CF). Clean plate A parts, test-fit the hull	plate B (about 2.5 h)
4:00	8b. Battery contacts and BMS wiring off the dart	plate B
6:00	8c. Drill arms, motors, sleeves (plate B done)	
7:00	8d. Populate the tray, battery power-up check	
8:30	8e. Close the hull, couplers, pods	
9:30	8f. Kick leg and foot	
10:15	10. Function test and calibration	
12:00	Done. Charge the cells (section 10, Charging).	

Bambu Studio settings (both plates): PLA, 0.4 nozzle, 0.16 mm layers, 4 wall loops (the 1.5 mm walls come out solid), 40% gyroid infill, top/bottom 5 layers, tree supports ONLY where the table says (inside the two hull halves), brim ON (auto) for the augers, X-Y hole compensation from the coupon (section 4). Couplers in PPA-CF (hardened nozzle, dried). Plate times above are estimates; Studio shows the real ones.

Part	Print pose	Supports	Size on bed mm	cm³
hull_top	split face down (on the bed), saddle up	tree supports ON, build plate only: the inner roof of the half shell (about 28 mm wide strip); pivot blocks, saddle and ear print without (arc outlines, 45 deg undersides)	121 x 55 x 35	18.77
hull_bottom	split face down, belly up	tree supports ON, build plate only: the inner roof of the half shell	121 x 42 x 21	11.4
nose	tip down on the bed (flat Ø22), spigot up	none (tip down, cone walls under 45 deg)	42 x 42 x 27	8.21
tail_cap	end face down, spigot up (no supports)	none (end face down; the socket windows bridge 8 mm)	42 x 42 x 15	8.03
tray	keel down on the bed; discs stand up, decks bridge 26 mm between discs	none (decks bridge 26 mm between discs; enable 'thick bridges')	109 x 39 x 36	14.91
kick_leg	on its side (pole axis horizontal, hub face down)	none (hub face and foot housing outboard face on the bed)	121 x 28 x 15	8.5
drill_leg_L	standing on the head's flat front face (local -x down), no supports	none (flat-bottomed head, D-bore roof bridges 5 mm)	19 x 21 x 33	3.66
drill_leg_R	standing on the head's flat front face (local -x down), no supports	none (flat-bottomed head, D-bore roof bridges 5 mm)	19 x 21 x 33	3.66
drill_leg_L_auger	standing on its base, brim 5 mm	none (45 deg lower flight flank); brim 8 mm, the base is small (flat core end only)	22 x 22 x 100	11.37
drill_leg_R_auger	standing on its base, brim 5 mm	none (45 deg lower flight flank); brim 5 mm, the base is small	22 x 22 x 100	11.37
foot_auger	base down	none; brim 5 mm	13 x 13 x 68	3.52
coupler_L	lying on its D flat (layers along the axle), PPA-CF	PPA-CF (hardened nozzle, dried filament), lying on the D flat, no supports; PLA only as a stopgap	7 x 29 x 6	0.98
coupler_R	lying on its D flat (layers along the axle), PPA-CF	PPA-CF (hardened nozzle, dried filament), lying on the D flat, no supports; PLA only as a stopgap	7 x 28 x 6	0.93
pod_cap_L	flat on the bed	none	10 x 10 x 2	0.1
pod_cap_R	flat on the bed	none	10 x 10 x 2	0.1
tail_boom	lying on its side (layers along the boom), PPA-CF	PPA-CF, lying on its side, no supports (the shank bridges 12 mm)	43 x 12 x 10	2.75
boom_anchor	flat on the bed, PPA-CF	PPA-CF, flat, no supports	31 x 32 x 4	2.03
drill_leg_L_tip	spigot up	none (spigot down)	15 x 15 x 10	0.89
drill_leg_R_tip	spigot up	none (spigot down)	15 x 15 x 10	0.89

4. Fit coupon (0:00, 20 minutes)

Print fit_coupon.stl. It has, left to right: the worm N20 pocket, the drill motor pocket, the F693ZZ seat with its flange recess, the hull boss bore, the N20 D-bore, an M3 pilot, an M3 clearance hole, the pod D-bore, the 3.8 shaft clearance, a coupler peg, and a hull ring with its spigot ring. Second row (v31): the JST-XH 2-pin and 3-pin header seats of the tail cap, the USB-C window of the hull (your cable must plug through it), the pod head D-bore with its cap counterbore, and a pod cap.

Check	Should feel	If not
Worm N20 into the square pocket	Slides in by hand, no wobble	Tight: hole compensation +0.05. Loose: -0.05.
Drill motor into its pocket	Slides in by hand	Same as above
F693ZZ into the seat	Firm thumb press, flange flush in the recess	Too tight: +0.05. Falls out: a drop of CA on assembly day.
Coupler peg into the boss bore	Turns freely, little play	Tight: sand the peg lightly
Coupler peg D-socket onto an N20 shaft	Pushes on firmly, no rotation slop	Loose: a drop of CA on assembly
Coupler peg into the pod D-bore	Slides in, no rotation slop	
M3 screw into the 2.5 pilot	Cuts its own thread, holds	
Spigot ring into the hull ring	Push fit, no rattle	Loose: one layer of tape on the spigots
JST headers into their seats	Push in, sit on the floor lip, no wobble	Loose: a drop of CA on assembly day
USB-C cable through the window	The overmold passes the window	Too tight: file the window corners on the hull
Pod cap in the counterbore	Drops in, turns freely	

Directions used below: left = the kick-leg side, seen from the tail with the saddle on top (y plus in the CAD). The left pod is the brighter red one in the viewer.

5. Start plate A and flash the firmware (0:30)

1. Send plate_A_hull.3mf to the X2D.
2. In the Arduino IDE: board XIAO_ESP32S3, Tools > USB CDC On Boot: Enabled (the sketch will not compile otherwise). Library Manager: install Adafruit INA219 and Adafruit BNO08x.
3. Nothing else connected to the XIAO. Plug it in by USB-C, open firmware/kedge_dart/kedge_dart.ino, Upload. Re-select the board if the IDE switches it to the wrong one.
4. Serial Monitor at 115200, 'No line ending'. You should see the KEDGE ready help line (INA219 and BNO085 'not found' is normal now). If the monitor stays blank: press the XIAO reset button once, check the port and USB CDC On Boot.
5. On your phone or laptop, join WiFi KEDGE (password grappstar) and open 192.168.4.1: you get the control page. From now on you can also upload over WiFi: Arduino IDE > Port > the network port called kedge. Unplug USB.

6. Prepare boards and wires (1:00)

1. Clip every header pin flush on the XIAO and both DRV8833 boards (flush cutters, close to the board). Cut the INA219 screw terminal off. Tin each pad you will use.
2. Set the dart MP1584 to 5.04 V before it is wired in (WIRING.md step B2). Set the spare one to 8.0 V: your bench motor supply.
3. Cut all wires to the lengths in section 9 (route column: KEEL = the channel under the cells, TOP = over the VTX). Strip 3 mm, tin, tape flag with the wire number. Heat shrink on before the second end is soldered.

7. Solder the harness and test it (2:00)

Follow firmware/dart/WIRING.md, Path B, steps B1 to B10, and its power-on check, battery unplugged at every step. Then BENCH_TEST.md steps 0 to 6 on the loose motors (directions and free currents). v31 positions: XIAO on deck B turned with its USB-C to the RIGHT flank (the hull window), DRV#1 beside it; DRV#2 and INA219 on deck C; BNO085 in the sled under the MP1584 shelf, turned across the dart; power loop socket in the TOP seat of the tail cap, charge socket in the bottom seat. Drill motors: bend both tabs up before soldering.

8. Mechanical assembly

8b. Battery contacts and BMS, off the dart (4:00)

1. Join the two springs with wire 2 (the series link, 22 AWG) and solder wire 3 (BMS BM) to that link. Solder wire 1 to one flat plate and wire 4 to the other.
2. On the bench, solder wires 1, 3 and 4 to the BMS B-, BM and B+ pads, wire 13 to P-, wire 8 to P+, and wires 5 to 7 to B-, BM, B+ for the charge socket. Solder the JST-XH 3-pin header on wires 5 to 7 and the 2-pin header on wires 8 / 9.

8c. Drill arms, motors and sleeve augers (6:00, plate B done)

1. Solder 200 mm of lead to each drill motor first (tabs bent back), and feed the leads through the bushing's back wall and out of the arm neck.
2. Push the motor can into the arm's bushing, shaft first out, until the end cap is 1.2 mm off the back wall: the can's flats key it. A drop of CA on one flat holds it.
3. Drop an M3 nut into the side slot near the tip of the sleeve. Sleeves are handed: the LEFT arm takes drill_leg_L_auger (right-hand flight), the RIGHT arm takes drill_leg_R_auger (left-hand flight).

4. Slide the sleeve over the motor from the front and turn it: the shaft finds the nut and the sleeve screws on until its base is 0.2 mm off the arm's shoulder (it must still turn by hand). Threadlocker on the shaft end, then an M3 nyloc down the tip bore with a 5.5 nut driver, tight against the hub wall while you hold the sleeve: the jammed pair keys the sleeve to the shaft in both directions.
5. Check: the sleeve turns with the motor without rubbing the bushing or the motor (no tick per turn).

8d. Populate the tray (7:00)

1. Glue the joined springs into the long recess on the FRONT cell-bay disc (nose side), one spring in front of each cell position. Glue the wire 1 plate into the BACK disc recess on the right (cell 1 side) and the wire 4 plate on the left. Lay the BMS in the keel channel under the cell bay: the KEEL wires run beside it through the same channel.
2. Deploy N20 into the sled cradle (worm box toward the tail, output shaft across), zip tie through the groove under the cradle.
3. Stick the boards on with foam tape: BNO085 on its two rails in the sled (across the dart), MP1584 on the shelf above it; XIAO on deck B with its USB-C at the right-hand edge (it must line up with the hull window) and DRV#1 beside it; DRV#2 and INA219 on deck C; VTX on the deck over the cells; the XIAO FPC antenna on the inside of the hull top over the board bay. KEEL wires: down the slot behind the XIAO, along the keel channel under the cells, out behind the deploy motor. TOP wires: through the top notches over the VTX. VTX antenna along the top through the disc 4 notch.
4. Push the 2-pin header into the TOP seat of the tail cap from the inside (pins down through the floor opening) and the 3-pin header into the bottom seat; a drop of CA each. Tape over the charge socket now: its pins are live as soon as the cells go in.
5. Cells in: cell 1 (right) button toward the nose, cell 2 (left) button toward the tail. Each pushes in sideways, squeezing its spring, and sits hard against its back plate.
5a. Loop still out. Meter on the charge socket pins (lift the tape corner): pin 1 to 2 reads 3.6 to 4.2 V, pin 2 to 3 reads 3.6 to 4.2 V, pin 1 to 3 reads 7.2 to 8.4 V. Any negative reading: pull the cells out and check their direction.
6. Battery power-up check: check the u.FL antenna is still clicked onto the VTX, then plug the loop in. WiFi page: i reads 7.4 to 8.4 V; z finds 0x40 and 0x4a; the deploy N20 runs on 1. Pull the loop plug.

8e. Close the hull (8:30)

1. Kick N20: its RIGHT stub cut to 1 mm past the worm box (wrap the motor so no swarf gets in). Kick leg first goes on the kick N20, off the dart: push the hub onto the LEFT stub (D flat); the motor sits beside the leg on the right. Lower motor and leg together: the motor into the saddle on the right of the centreline, the left stub down the vertical slot in the saddle's inner wall, worm box toward the nose; the leg lies on the centreline. The square worm box keys the motor in its square pocket: a drop of CA on the can stops it rattling. Leads down through the hole in the saddle floor right behind the end cap.
2. Camera into the nose pocket from the back (lens into the hole), cable out the back. Tray into the bottom half, and seat the nose spigot in the bottom half at the same time: the groove on the spigot sits over the half-rib inside the hull.
3. Thread the leads before the top half goes on: kick leads (wires 43, 44) up through the saddle floor hole; foot leads (49, 50) out through the skin hole under the folded pole; drill leads (45 to 48) out through the two holes just behind the pivot blocks. Lower the top half on.
4. Tail boom: push its square shank through the anchor disc (head into the recess) and through the tail cap's square hole from inside; fit the tail cap with the disc in its spigot. Radial screws: four M3 x 8 into disc 1 (z ±3 both sides) and four M3 x 8 through the tail spigot into the anchor disc: those four carry the tow. The top half's rib closes the nose groove, so the nose is locked. Tether: a loop round the M3 x 8 pin in the boom eye.
5. Solder the kick N20 and the two drill motors to their leads (as marked in 7.6), heat shrink (slide it on before soldering; iron on each joint 3 s at most; shrink with a heat gun on low, no lighter near the PLA).
6. Couplers: slide coupler_L in through the left boss, coupler_R through the right, turning gently until each D-socket drops onto the deploy N20 stub. Drill arms onto the coupler ends (D-bore), printed pod cap into each head's outer counterbore, M3 x 8 through the cap into the coupler. The cap sits on the coupler end: the arm must still swing freely (0.7 mm float). Fold the sleeves along the flanks: the heads rest just clear of the fold stops on the pivot blocks; swung down they land on the deploy stops in front of the heads.

8f. Kick leg and foot (9:30)

1. Press the F693ZZ into the ear (left of the hub) from outside, flange down in the counterbore: its inner face stands 0.4 proud on the hub side and touches the hub's small boss.
2. Foot N20 into the foot housing from the open inboard side, output shaft sliding along the slot into the top hole, strap. Foot auger onto the shaft (it sits 0.3 above the housing), grub screw onto the flat.
3. Feed the foot leads from the foot pocket through the pole channel and out the small hole near the hub. Solder them to wires 52 and 53 (as marked in 7.6): heat shrink on first, iron 3 s at most per joint, heat gun on low, no lighter near the PLA. Leave the loop round the hub.

4. M3 x 8 through the ear bearing into the hub (no washer: the head clamps the inner race onto the hub boss). Plug the loop in (top of the tail cap) and send @ until the leg rests folded on the hull.

10. Function test and calibration (10:15)

Use a tray of damp soil or sand. Laptop on the KEDGE WiFi page; hold a key to run, let go to stop. Space bar stops everything.

Key	Should happen	If it goes the wrong way
First, in the air	Hold each key for half a second: directions were fixed in 7.6	If one is still wrong, swap it at the DRV end (open the hull)
2 (hold)	Kick leg swings up over the nose and down; the foot auger tip meets the soil near 170°	Swap wires 43 and 44
2 and 4 (hold both)	Foot auger screws IN while the kick keeps pushing, until the foot housing sits on the soil and the foot auger stands vertical (on a hard floor the pole lands in its flute on the nose top: that is the kick rest)	Foot backs out: swap wires 49 and 50
1 and 3 (hold both)	Both pods swing down and back while BOTH augers spin and mill in; the dart does not twist; in air the arm heads stop on the deploy stops of the pivot blocks	Pods go up: swap 41 and 42. One auger backs out: swap that motor's own pair on AOUT1/AOUT2. Both: swap 37 and 38 at the XIAO
i during each	Note the free current, the current when the foot touches, and the stall current	Put them in kedge_logic.h (kickContactMa, kickStallMa, deployStallMa) and upload over WiFi
b	Roll and pitch of the planted dart	
a	Full auto sequence, same order as above, by itself	Space aborts
A then throw	Arms auto-anchor on landing (the dart anchors 1.5 s after it comes to rest)	Press A again to disarm

Charging: either take the cells out and use the XTAR MC2, or plug a 2S balance charger (JST-XH 3-pin) into the charge socket under the tail. Loop plug (top of the tail) OUT while charging. Firmware updates: OTA over WiFi, or USB-C through the window in the right flank with the loop plug out.

Recovery: # (drills out), ! (pods up), $ (foot out), @ (kick back). The worms hold every position with the power off.

Note: STOP (space, /stop, or a client hold ending) also aborts a running auto sequence; that is deliberate.

Pins (firmware v4): drills on DRV#2 A (D8/D9), foot on DRV#2 B (D6/D7). D6/D7 flicker at every reset; the slow foot worm does not care.

9. Wire cut list (pad to pad)

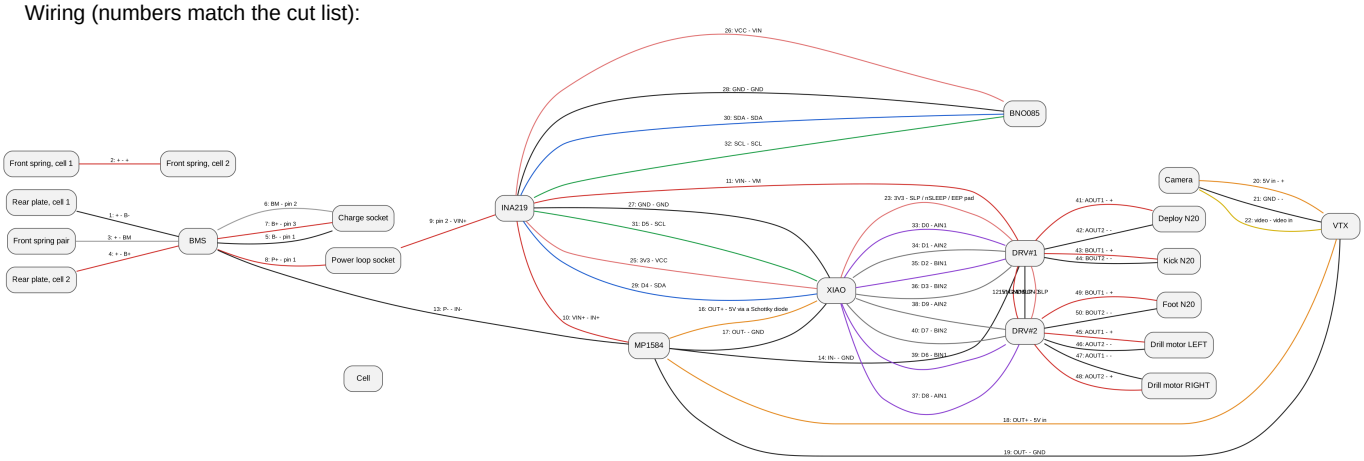
#	From	To	Colour	Gauge	Cut mm	Note
1	Rear plate, cell 1 (-)	BMS B-	black	22 AWG	45	solder to the plate tab before it goes in its recess
2	Front spring, cell 1 (+)	Front spring, cell 2 (-)	red	22 AWG	45	the series link: full battery current; glue both springs in the front recess
3	Front spring pair (cell 1 + / cell 2 -)	BMS BM	white	26 AWG	55	midpoint sense
4	Rear plate, cell 2 (+)	BMS B+	red	22 AWG	45	
5	BMS B-	Charge socket pin 1 (tail cap, bottom seat)	black	26 AWG	90	JST-XH 3-pin, balance charge; along the keel channel
6	BMS BM	Charge socket pin 2 (tail cap, bottom seat)	white	26 AWG	90	
7	BMS B+	Charge socket pin 3 (tail cap, bottom seat)	red	26 AWG	90	
8	BMS P+	Power loop socket pin 1 (tail cap, top seat)	red	22 AWG	105	JST-XH 2-pin on top of the tail cap; the shorting plug is the ON switch
9	Power loop socket pin 2	INA219 VIN+	red	22 AWG	80	main + after the loop
10	INA219 VIN+ (same pad)	MP1584 IN+	red	22 AWG	200	buck taps before the shunt; keel channel to the sled
11	INA219 VIN-	DRV#1 VM	red	22 AWG	50	motor rail 6.0-8.4 V
12	DRV#1 VM	DRV#2 VM	red	22 AWG	70	daisy chain ends here (no DRV#3)
13	BMS P-	MP1584 IN-	black	22 AWG	90	main ground
14	MP1584 IN- (same pad)	DRV#1 GND	black	22 AWG	200	
15	DRV#1 GND	DRV#2 GND	black	22 AWG	70	
16	MP1584 OUT+ (5.04 V)	XIAO 5V via a Schottky diode (stripe toward the XIAO)	orange	24 AWG	195	set and check 5.04 V BEFORE connecting; 1N5817 or SS14 in heat shrink
17	MP1584 OUT-	XIAO GND	black	24 AWG	195	
18	MP1584 OUT+	VTX 5V in	orange	26 AWG	95	up beside the deploy motor, over the top
19	MP1584 OUT-	VTX GND	black	26 AWG	95	
20	VTX 5V in (same pad)	Camera + (red)	orange	28 AWG	120	forward along the top through the disc notches
21	VTX GND (same pad)	Camera - (black)	black	28 AWG	120	
22	Camera video (yellow)	VTX video in	yellow	28 AWG	120	twist with the camera ground
23	XIAO 3V3	DRV#1 SLP / nSLEEP / EEP pad (NOT VCC or VIN)	red/white	28 AWG	55	wakes the driver; meter check first (WIRING.md step B6)
24	DRV#1 SLP	DRV#2 SLP	red/white	28 AWG	70	
25	XIAO 3V3	INA219 VCC	red/white	28 AWG	60	
26	INA219 VCC	BNO085 VIN	red/white	28 AWG	180	the BNO085 is in the sled now: keel channel
27	XIAO GND	INA219 GND	black	28 AWG	60	
28	INA219 GND	BNO085 GND	black	28 AWG	180	
29	XIAO D4	INA219 SDA	blue	28 AWG	60	I2C SDA
30	INA219 SDA	BNO085 SDA	blue	28 AWG	180	twist SDA with GND along the keel
31	XIAO D5	INA219 SCL	green	28 AWG	60	I2C SCL
32	INA219 SCL	BNO085 SCL	green	28 AWG	180	
33	XIAO D0	DRV#1 AIN1	purple	28 AWG	55	deploy
34	XIAO D1	DRV#1 AIN2	grey	28 AWG	55	deploy
35	XIAO D2	DRV#1 BIN1	purple	28 AWG	55	kick
36	XIAO D3	DRV#1 BIN2	grey	28 AWG	55	kick
37	XIAO D8	DRV#2 AIN1 (only AIN1: NO bridge to BIN1)	purple	28 AWG	50	drills
38	XIAO D9	DRV#2 AIN2 (only AIN2: NO bridge to BIN2)	grey	28 AWG	50	drills
39	XIAO D6	DRV#2 BIN1	purple	28 AWG	50	foot (slow worm, harmless if it twitches at reset)
40	XIAO D7	DRV#2 BIN2	grey	28 AWG	50	foot
41	DRV#1 AOUT1	Deploy N20 + (red)	red	26 AWG	220	keel channel, along the keel under the BNO085, up the disc 1 notch to the end-cap tabs
42	DRV#1 AOUT2	Deploy N20 - (black)	black	26 AWG	220	

#	From	To	Colour	Gauge	Cut mm	Note
43	DRV#1 BOUT1	Kick N20 + (red)	red	26 AWG	215	keel channel, up behind the deploy motor, through the saddle floor hole to the end-cap tabs
44	DRV#1 BOUT2	Kick N20 - (black)	black	26 AWG	215	
45	DRV#2 AOUT1	Drill motor LEFT + (red)	red	26 AWG	170	over the top, out the left hole behind the boss, 25 mm loop into the pod top at the end cap
46	DRV#2 AOUT2	Drill motor LEFT - (black)	black	26 AWG	170	
47	DRV#2 AOUT1 (same pad as the LEFT red)	Drill motor RIGHT - (black) SWAPPED	black	26 AWG	170	reversed on purpose: the pair counter-rotates
48	DRV#2 AOUT2 (same pad as the LEFT black)	Drill motor RIGHT + (red) SWAPPED	red	26 AWG	170	
49	DRV#2 BOUT1	Foot N20 + (red)	red	26 AWG	300	over the top, out the skin hole at x 42, 60 mm loop round the hub, through the pole channel
50	DRV#2 BOUT2	Foot N20 - (black)	black	26 AWG	300	silicone wire, it flexes every deploy

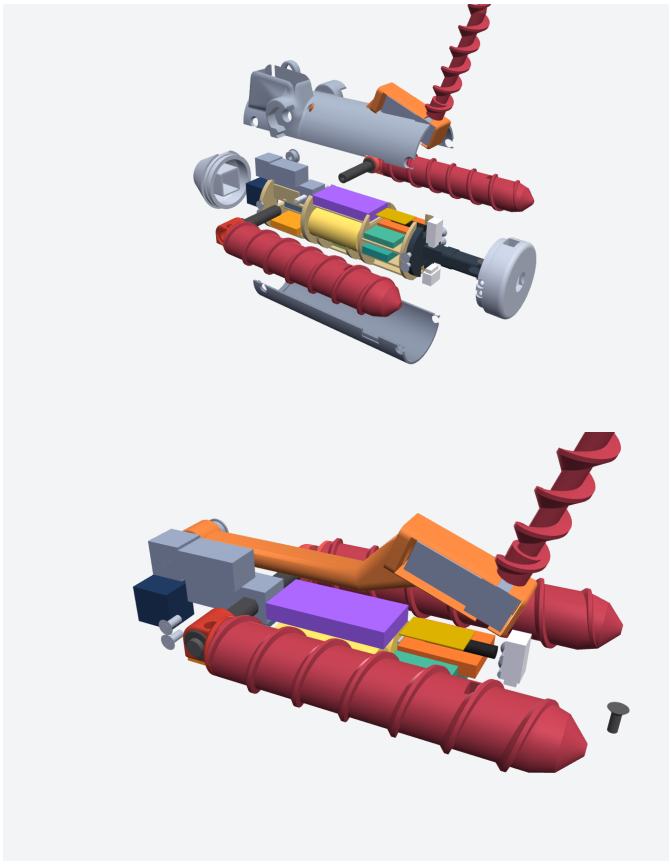
Totals by gauge and colour (mm): 22 AWG black 405, 22 AWG red 595, 24 AWG black 195, 24 AWG orange 195, 26 AWG black 1260, 26 AWG orange 95, 26 AWG red 1165, 26 AWG white 145, 28 AWG black 360, 28 AWG blue 240, 28 AWG green 240, 28 AWG grey 210, 28 AWG orange 120, 28 AWG purple 210, 28 AWG red/white 365, 28 AWG yellow 120

11. Pictures

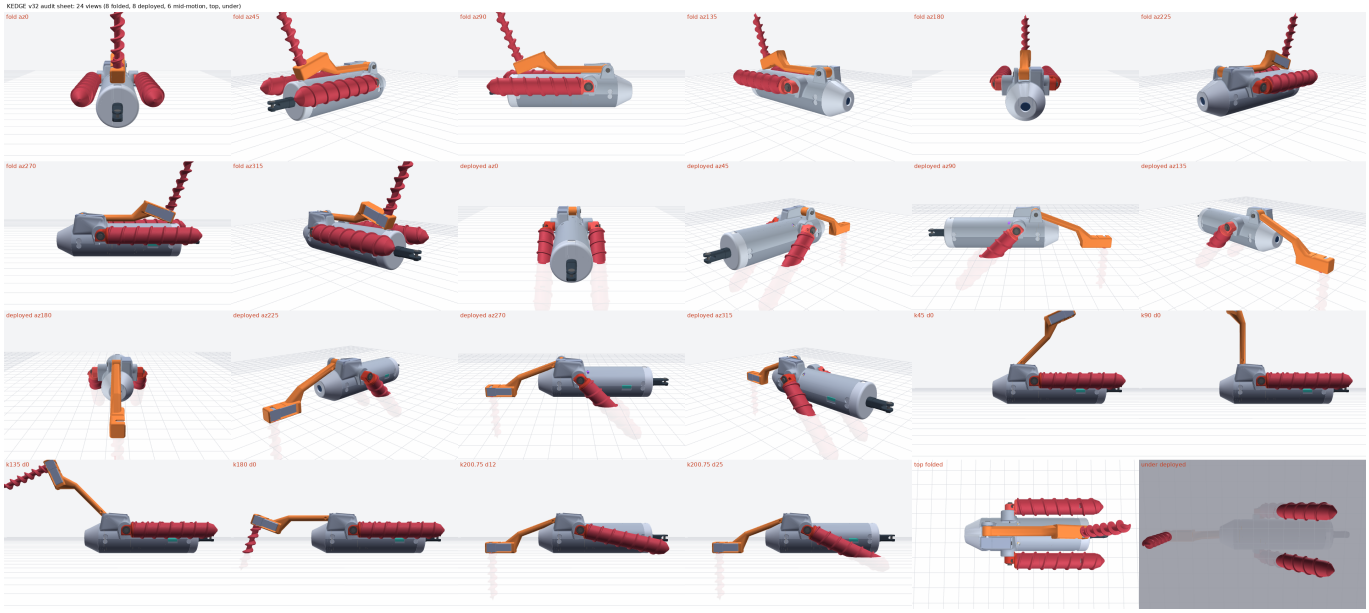
Wiring (numbers match the cut list):



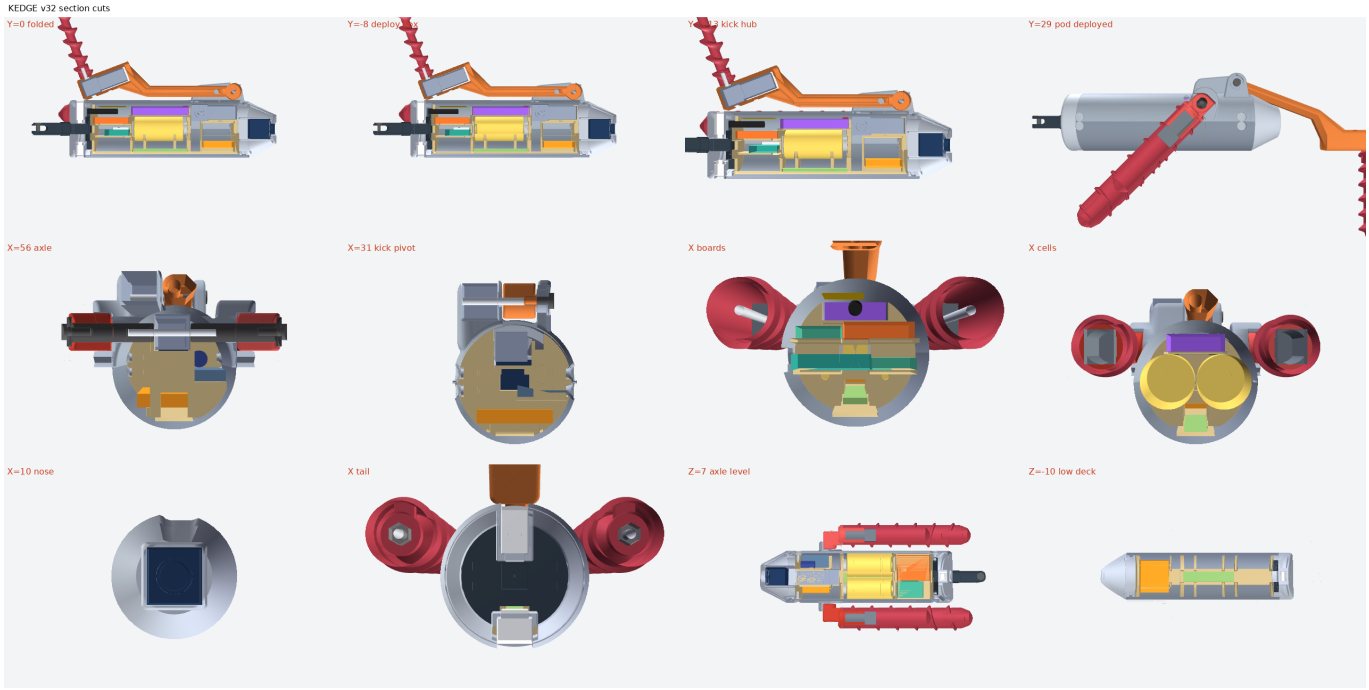
Exploded view and populated tray:



Audit sheet: 24 views of the exact print geometry (8 folded, 8 deployed, 6 mid-motion, top, under).



Section cuts:



12. Design notes

• Drills: each gearmotor is fixed in its arm and sits inside its sleeve; the sleeve runs on the arm's bushing and shoulder (push goes into the arm), the motor shaft turns it through the jammed nut pair at the tip.
• The drill pair counter-rotates so the dart does not spin; the two augers are mirror images so both screw in.
• The deploy N20's own shaft stubs drive the pods through two Ø7 D couplers (PPA-CF) that run in the pivot blocks. At full deploy the arm heads sit on the deploy stops carved in the pivot blocks: tow pull turns the pods further down, so that load goes into the hull, not into the worm or the couplers.
• The flute in the nose top does the same for the kick leg: tow lifts the nose, the soil pushes the planted foot down, the pole lands in the flute (the kick rest). The foot housing is tilted on the leg so the foot auger plants vertical.
• Tether load path: boom eye pin, PPA-CF tail boom, its head in the anchor disc, 4 radial M3 through the tail spigot into the hull. The line is a loop round the M3 x 8 pin in the boom eye.
• Ground geometry: kick stops at 200.75° with the foot housing on the ground and the foot auger 67.9 mm in; drills stop at 45.0° with the pods 8.91 mm clear and the augers 49.1 mm in.
• Power: 2S straight to the DRV8833 motor rail through the INA219; one MP1584 at 5.04 V for the XIAO, VTX and camera.
• Below the centreline the hull is a plain Ø42 cylinder for the launcher cradle; the folded pods sit just above it. Nothing is plugged into the belly: the ON plug is recessed in the top of the tail cap.